

D.Ekanth Reddy

System Engineer

Phone: 8618197429 | Email: dekanthreddy@gmail.com

PROFESSIONAL SUMMARY

- Embedded Software Engineer with around 3 years of experience focusing on C / C++11 / Embedded C.
- Expert in developing robust, efficient, and maintainable system-level software compliant with real-time and hardware-constrained platform requirements.
- Experienced in Qt framework development, utilizing QML for responsive UIs and Qt Widgets for complex data-centric applications.
- Competent in designing and implementing Linux-compatible Qt-based GUI applications with an emphasis on performance and usability.
- Extensive experience with ARM Cortex-A / 8-bit microcontrollers, specializing in peripheral interfacing and Embedded application development.
- Skilled in Embedded C programming with extensive GDB debugging, code optimization and resolving complex challenges.
- Implemented and debugged embedded protocols like SPI, I2C, UART, and TCP/IP for robust device communication and data transfer.
- Expertise in building high-performance, reliable Linux applications with multithreading, socket programming & IPC.
- Results-driven SQL database with experience in data organization, query optimization and database management to support data-driven decision making.
- Proficient in creating diagnostic reports, technical guides, and comprehensive software documentation to support maintenance and facilitate knowledge transfer.
- Effectively managed the complete embedded software development lifecycle within Agile/Scrum environments, ensuring timely and high-quality product delivery.

TECHNICAL TOOLS

- **Languages** : C, C++11, Embedded C, QML
- **Tools** : Qt Creator, Keil, SQLite
- **Processors** : ARM Cortex-A, 8-bit MCUs
- **Protocols** : UART, I2C, SPI, TCP/IP
- **Operating Systems** : Linux (Ubuntu), Windows

EXPERIENCE

- Worked as System Engineer in Web India Pvt Ltd from April 2025 to Jan 2026.
- Worked as an Engineer in SVN Technologies From May 2023 to March 2025.

PROJECTS

1. Smart Message Interaction Framework for Digital Interactions

Role : Developer

Environment : C++, QT Creator, ARM-CORTEX-A.

Description :

Developed a secure and scalable digital data exchange system prioritizing data integrity and reliability. Ensured seamless access to digital information across diverse applications. Designed for adaptability, supporting a broad range of use cases with robust security measures.

Responsibilities:

- Collaborate with cross-functional teams to ensure cohesive development and timely delivery of modules.
- Built maintainable embedded C++11 code and GUI components, emphasizing modularity and testability.
- Conducted debugging, troubleshooting, and system improvements to enhance reliability and efficiency.
- Designed and delivered scalable Qt socket-based data communication, enhancing overall application performance.

2. ARM based Intelligent Refreshment Management Platform

Role : Developer

Environment : ARM CORTEX-A / C++11/ Qt Creator /Linux

Description:

Developed an advanced system to automate and simplify formulation dispensing and mixing processes, eliminating manual formulations. It enhances precision, efficiency, and reliability with real-time tracking and seamless management. This upgrade ensures consistent quality, reduces delays, and optimizes overall production.

Responsibilities:

- Reviewed and interpreted system specifications to ensure accurate and efficient software design.
- Built and refined Qt-based application components to ensure compliance with functional and design specifications.
- Developed visually consistent HMI interfaces using standardized iconography to enhance usability.
- Integrated robust database functionality to ensure reliable data management and retrieval.
- Performed seamless module integration and rigorous testing to maintain system integrity and quality standards.

3. Advanced Contamination Monitoring and Alert System

Role : Programmer

Environment : 8-bit microcontroller, ADC, Keil, Embedded C, UART

Responsibilities:

- Developed microcontroller software using Embedded C to achieve reliable real-time data handling and system control.
- Integrated ADC for data collection, real-time signal processing, and automated control functionalities.
- Developed UART communication protocols to enable system monitoring and parameter configuration.
- Coordinated development tasks and delivered high-quality modules compliant with integration and testing requirements.

EDUCATIONAL QUALIFICATION

2019-2022 SV Degree College, **B.Com(Comp)**

PERSONAL DETAILS

Nationality : Indian

Marital Status : Unmarried

Language : English, Telugu &Hindi